

Are Bikeshare Users Different from Regular Cyclists?

A First Look at Short-Term Users, Annual Members, and Area Cyclists in the Washington, D.C., Region

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In recent years, bikeshare systems have spread throughout the United States. Despite the increase in systems and users, little is known about how bikeshare member demographics and travel patterns compare with those of traditional bicyclists. To bridge the gap, this study investigated bikeshare system user travel behavior and developed a profile of user demographics through the comparison of short-term (1 day) users and annual members of Capital Bikeshare (CaBi) in Washington, D.C., with profiles of area bicyclists. Data originated from a Washington, D.C., regional household travel survey of 2007–2008, an intercept survey of short-term CaBi users, and an online survey of annual CaBi members. The analysis compared gender, race and ethnicity, age, and student status, as well as socioeconomic status across the surveys. The comparison included trip purpose, helmet use, and travel modes for trips that were replaced by bikesharing. Profiles of CaBi and area cyclists differed with respect to many demographic and socioeconomic characteristics. Compared with area cyclists, CaBi short-term users and annual members were more likely to be female and younger, to have lower household incomes, and to own fewer cars and fewer bicycles and were more likely to cycle for utilitarian purposes. CaBi trips mainly replaced trips made by public transport and by walking. CaBi short-term users and members shared similar characteristics, but short-term users were more likely to ride for recreational purposes and were less likely to wear a helmet. The study results indicated that bikesharing can encourage new segments of society to cycle and can help increase the bicycling mode share overall.

Over the past 5 years, U.S. cities such as Minneapolis, Minnesota; Denver, Colorado; Boston, Massachusetts; and Washington, D.C., have implemented third-generation bikesharing systems. These new bikeshare systems have overcome many of the problems associated with older bikeshare programs, which relied on free or coin-deposit payment for bicycles. Third-generation bikesharing systems have docking stations and electronic kiosks to rent and return bicycles. They provide key cards for system access and have a fleet of specially designed, colored, and branded bicycles (1, 2).

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Most of the research on bikesharing in the United States has focused on the evolution of bikeshare systems over time, ridership trends, successful implementation strategies for new systems, and the potential benefits of bikesharing (3–6). Little is known about bikeshare user demographics, rider motivation, travel behavior, and similarities and differences between annual bikeshare members, short-term (1 day) users, and regular cyclists. This paper presents a case study of short-term and annual bikeshare users of Capital Bikeshare (CaBi) in Washington, D.C., and Arlington County, Virginia. At the time of this writing, CaBi was the largest bikesharing program in the United States, with 1,300 bikes and 144 stations in Washington, D.C., and Arlington County (7). The system was expected to expand to the City of Alexandria, Virginia, and to Montgomery County, Maryland, by the end of 2012.

The goal of the study reported here was to provide a profile of short-term and annual CaBi users and their travel behavior compared with that of traditional cyclists in the Washington, D.C., area. Data for the comparison of short-term CaBi users and annual CaBi members originated from a 2011 intercept survey of short-term CaBi users (8) and original analysis of CaBi's 2011 online survey of annual members (9). Demographics for short-term and annual CaBi users were compared with those of Washington, D.C., area cyclists, and data were analyzed from the Metropolitan Washington Council of Governments (MWCOC) 2007–2008 regional travel survey (10).

The next sections of this paper briefly introduce the concept of bikesharing, provide a short history and ridership trends for CaBi, and an overview of the existing literature on characteristics of bikeshare users. After a short description of the three data sets, an analysis is presented that compares Washington, D.C., area cyclists with short-term and annual CaBi users.

BACKGROUND

This section highlights the history and reported benefits of bikesharing in the United States. More background information can be found in the detailed overviews about the history of bikesharing worldwide by Shaheen et al. (1, 6) and DeMaio (2). The results also are briefly summarized from some exploratory studies, including those of Shaheen et al. (1, 6) and DeMaio (2), which evaluated the alleged benefits of bikesharing. Finally, this paper briefly summarizes the only published academic report on the demographics of annual bikeshare members in the United States (6). To date, no study about bikesharing in the United States has distinguished between short-term

users and annual members, and no study has compared those two groups of bikeshare users with regular cyclists.

Bikesharing originated in Amsterdam in the Netherlands with the White Bikes program in 1965 (*1*). This first-generation bikesharing system consisted of brightly colored bicycles left unlocked in Amsterdam free of charge for anyone to use. Damage and theft of bicycles led to the decline of first-generation bikeshare programs. Coin-deposit-operated systems were considered the second-generation of bikesharing. The second and subsequent generations continued to employ the uniform color scheme pioneered by first-generation systems. Second-generation systems also featured docking stations and a mechanism that required a refundable deposit to unlock a bicycle. However, the anonymity of system users and the inexpensiveness of the deposits failed to prevent bicycle theft (*1, 2, 6*).

Third-generation bikeshare systems have attempted to overcome many of the problems associated with the older systems. Technological advances have allowed bikesharing providers to use telecommunication systems. Smartcards and key fobs allow users to electronically lock and unlock bicycles from kiosks (docking stations). Smartcards and fobs include identifiable links to users, which eliminate the anonymity of bikeshare users. System operators use telecommunications to collect data on usage, such as duration of checkout and availability of bicycles at docking stations. Users can make reservations and check for bicycle availability through mobile phone applications (*1, 2, 6*). In Europe, the deployment and expansion of third-generation bikesharing schemes in recent years have been rapid and such schemes have by now approached market saturation (*11*).

The reported benefits of bikesharing include a reduction in greenhouse gas emissions through a modal shift from automobiles to bicycles and reduced fuel consumption; lower household transportation expenditures; increased accessibility to public transport through a resolution of the first mile–last mile problem; and increased physical activity overall (*1, 2, 6*). To date, much bikesharing research has focused on the determinants of ridership [e.g., infrastructure, the built environment (*3–5, 12*)].

Several other studies of bikeshare systems outside the United States have provided information about annual members. Bachand-Marleau et al. collected basic demographic information and surveyed the attitudes of riders of the Bixi bikeshare system in Montreal, Quebec, Canada (*13*). Tang et al. collected information on bikeshare users in three cities in China (*14*).

As mentioned earlier, only one published academic study has provided demographic and socioeconomic details about bikeshare users in the United States (*6*), and no study to date has distinguished between short-term and annual members. Shaheen and colleagues conducted a survey of annual members of bikeshare systems in Minneapolis, Montreal, Denver, and Washington, D.C. Aggregate statistics for all four systems indicated that most bikeshare users were younger than 34 years of age (60%), white (approximately 80%), and highly educated (85% had a bachelor's degree or higher). The results on trip purposes were not entirely comparable across the cities surveyed, but to go to work or to school was the number one trip purpose in all systems.

CaBi OVERVIEW

Third-generation bikesharing in the United States was pioneered in Washington, D.C. The District of Columbia initiated a public–private partnership program called SmartBike D.C., operated by

Clear Channel Outdoor. The system operated from 2008 to early 2011 and consisted of 10 stations and 120 bicycles at implementation (*1*). However, the program ended as a result of financial and technological constraints including low ridership, an inability to expand the system, and the need for electrical connections at stations. Lessons learned from SmartBike D.C. paved the way for CaBi (*15*).

CaBi began operations September 20, 2010, with 1,100 bikes and 114 stations throughout the District of Columbia and Arlington County, Virginia (*16*). By February 2012, the system had expanded to 1,300 bikes and 144 stations (*7*). Plans to expand the system to the City of Alexandria, Virginia, and Montgomery County, Maryland, by the end of 2012 could increase the system to 2,800 bikes and 288 stations (*17*).

Users can join CaBi in two primary ways: (*a*) as an annual or a monthly member or (*b*) as a short-term member. Annual or monthly members pay an up-front fee (\$75 per year, \$25 for 1 month, or \$7 per month for 1 year) and register their name, address, and credit card online with CaBi. Fees for each bicycle checkout are billed to the credit card on file every month, with no charge for rides less than 30 min. Short-term members join through the purchase of 24-h (\$7) or 3-day (\$15) passes with a credit card at the bikesharing kiosks.

Both annual and short-term memberships have grown significantly. For example, there were more than 15,000 annual CaBi members in June 2011. By June 2012, annual membership had increased 57.6%, to more than 24,000. The number of short-term memberships almost tripled: from 55,717 in June 2011 to 169,635 in June 2012 (*15*).

The 202,822 trips systemwide in June 2012 represented an increase of 59,000 checkouts, or 41%, compared with 143,511 checkouts in June 2011. Even after an expected weather-related decrease in trips during the 2011–2012 winter months, March 2012 experienced 21,000 (or 15%) more checkouts than the highest recorded monthly trip count in 2011: 143,511 trips in June 2011 (*15*).

DATA SOURCES AND METHODS

On the basis of three travel surveys, the study reported here compared demographic, socioeconomic, and travel characteristics of area cyclists, short-term CaBi users, and annual CaBi members. Data for the analysis originated from the MWCOG 2007–2008 regional household travel survey of area cyclists, the 2011 Capital Bikeshare Casual User Survey, and the 2011 Capital Bikeshare Member Survey. The surveys varied in their objectives, methodology, and number of participants.

The Capital Bikeshare 2011 Member Survey was administered online to registered (i.e., annual and monthly) CaBi members over a 1-month period during November and December of 2011. The objectives of the survey were to ascertain demographic and use characteristics of CaBi members, satisfaction with the system, and changes in travel patterns on the basis of bikeshare availability. All registered CaBi members were notified of the survey by e-mail. The survey was completed by 5,464 members, which represented 31% of the total CaBi membership at that time. Although the survey was sent to annual and 30-day members, 30-day members represented only approximately 10% of all CaBi members at the time (*15*). Unfortunately, for purposes of this analysis, the data set did not allow for differentiation between responses from the two membership groups. The respondents to this survey therefore are referred to as annual members in this paper for the sake of brevity.

The 2011 Capital Bikeshare Casual User Survey was designed to understand the demographic profile of cyclists that had purchased

short-term CaBi memberships. The term “short-term members” refers to users who purchase a 24-h or 5-day membership (5-day memberships have been discontinued), rather than an annual or a monthly membership that requires a key fob to access the system. The intercept survey was designed and conducted by a graduate studio class from Virginia Polytechnic Institute and State University, Blacksburg, in collaboration with CaBi. The survey consisted of 23 questions. Surveys were conducted by graduate students across two weekends in September and October 2011 at five CaBi stations selected for high levels of use by short-term users. Short-term users that did not have time to complete the verbal survey were provided with instructions on how to complete the survey online. Surveys were completed by 340 short-term users, of whom 66% identified themselves as visitors from outside the Washington, D.C., Metropolitan Statistical Area. The intercept survey included responses from only one person if cyclists traveled in groups. Thus data were not collected for other bicyclists that accompanied a CaBi survey respondent.

The MWCOG regional household travel survey of 2007–2008 included 25,197 respondents, which included 832 cyclists that reportedly rode at least once a week. The objective of the survey was to evaluate the metropolitan Washington area transportation system’s effectiveness to meet local household transportation needs. Survey participants were selected by geographically stratified household random sampling. Randomly selected participating households kept a 1-day travel diary between February 2007 and May 2008. Each participating household member recorded their travel for 1 weekday, and transmitted it to the survey team through a computer-aided telephone interview. Data collected included trip purpose, mode of transport, distance, and trip duration. The MWCOG survey was conducted before the implementation of CaBi and provided a profile of area cyclists before the launch of bikesharing.

To mirror the main geographic area of CaBi use, the analysis reported here used only the responses of 285 D.C. area cyclists, who lived in the urban core of the region including Washington, D.C., and Arlington County and Alexandria City, Virginia. The term “D.C. area cyclists” refers to respondents in these three jurisdictions, who reported cycling at least once a week. Because the sample size was small, a separate analysis was conducted of cyclists for the entire region (with 832 cyclists) and for Washington, D.C., and the adjacent inner ring suburban jurisdictions of Arlington County, Alexandria, and Fairfax County in Virginia, and of Montgomery and Prince George’s Counties in Maryland (632 cyclists). Comparisons of regular cyclists and CaBi users yielded consistent results for cyclists in the entire region; cyclists in Washington, D.C., and the inner suburbs; and Washington, D.C., Arlington, and Alexandria only. In the section that follows, the results are presented for Washington, D.C., Arlington, and Alexandria only, because these areas constituted the most appropriate geographic scope for a comparison with CaBi users.

The next section also presents demographic estimates, and socioeconomic and trip purpose characteristics of area cyclists, short-term CaBi users, and annual CaBi members. Information on the statistical significance of differences between the groups of cyclists is provided in Table 1. An alternative approach to significance tests of binary data (other than chi-square) is to use *t*-tests that compare population proportions for each subcategory (e.g., proportion of women). Such tests were conducted. The chi-square tests for the binary data presented in Table 1 and the *t*-tests of population proportions provided consistent results. Results of the population proportion *t*-tests are available on request.

ANALYSIS AND DISCUSSION

Demographics

The demographics of CaBi members, CaBi short-term users, and D.C. area bicyclists are shown in Figure 1, and tests of significance are shown in Table 1. (For all figures, see Table 1 for detailed point estimates, 95% confidence intervals, and statistical tests of significance.) The reported gender of D.C. area bicyclists was 65% male and 35% female, which reflected the general percentage distribution of bicyclists in the United States. The CaBi intercept survey of short-term users observed that 48% of riders were male, and 52% were female. The high share of females among short-term users may be explained by the interview method used in this study. Interviewers, for the most part, interviewed only one respondent if short-term users traveled in groups. Thus women might have been more likely than men to act as respondents.

However, a higher share of women among short-term CaBi users was confirmed by the annual member survey. Respondents to the annual CaBi member survey were 55% male and 45% female. The data on gender were collected with respect to membership status only and did not reflect actual rides taken by respondents. More detailed analysis of the annual membership data indicated that women were more likely than men to make no or few trips per month: 21% of women reportedly made no CaBi trip in a typical month. Another 59% of women reportedly made between 1 and 10 trips per month. Only 20% of women made more than 10 CaBi trips per month. By contrast, only 13% of men reportedly made no CaBi trip per month, and 35% reportedly made 10 or more trips a month.

About 80% of D.C. area cyclists and CaBi users were white. Hispanics accounted for 3% to 5% of D.C. area cyclists and CaBi users. The most significant variations between D.C. area bicyclists and CaBi users were among Asians and African-Americans. Asians accounted for only 3% of area cyclists but made up 7% of CaBi members and 8% of CaBi short-term users. African-Americans accounted for almost 8% of area cyclists but only 3% of CaBi members and 5% of CaBi short-term users.

The study’s comparison showed that D.C. area cyclists were older than CaBi users. The reported age of D.C. area bicyclists showed a plurality of 31% in the range of 35 to 44 years old. In fact, 71% of D.C. area cyclists were older than 35. By contrast, a plurality of both CaBi short-term users (43%) and CaBi annual members (55%) fell within the range of 25 to 34 years old. The self-reported average age for area cyclists was 42, compared with 34 for CaBi members and 35 for short-term CaBi users. No significant differences were noted between D.C. area bicyclists and CaBi annual members in terms of their student status. Data about the student status of short-term CaBi users were not available.

Income, Car Ownership, and Access to a Bicycle

As shown in Figure 2, the data suggested that D.C. area cyclists had higher incomes than annual CaBi members. Significant differences, as shown in Table 1, were noted at the lower and upper ends of the income categories reported by D.C. area bicyclists and CaBi annual members: 45% of CaBi annual members reported an annual household income below \$75,000, while only 31% of D.C. area bicyclists did. At the upper range of household income, 53% of D.C. area bicyclists reported household incomes above \$100,000, while only 39% of CaBi members did. The mean household income for area cyclists

TABLE 1 Point Estimates, 95% Confidence Intervals, and Significance Test

							Differences Significant at $P < .05$ (test)	
	Area Cyclists		Short-Term CaBi		CaBi Members			
Variable	Percentage	95% CI	Percentage	95% CI	Percentage	95% CI	Area Cyclists vs. Both Groups of CaBi Users	CaBi Short-Term Users vs. Members
Gender–Sex								
Male	65.4	58.8–71.4	48.4	48.0–48.7	55.0	53.6–56.3	} Yes (χ^2)	No (χ^2)
Female	34.6	28.6–41.2	51.6	46.3–57.0	45.0	43.7–46.4		
Race–Ethnicity								
Asian	3.4	1.8–6.5	7.7	5.3–11.0	6.7	6.0–7.4	} Yes (χ^2)	No (χ^2)
African-American	7.8	4.9–12.3	5.0	3.1–7.9	3.4	2.9–3.9		
White	81.4	75.7–86.1	78.2	73.4–82.3	80.6	79.4–81.6		
Hispanic	3.4	1.5–7.6	4.4	2.7–7.2	4.9	4.4–5.6		
Other	4.4	4.0–5.1	3.0	1.4–5.0	4.5	4.0–5.1		
Age (years)								
Mean	42.0	40.5–43.45	34.8	33.5–36.0	34.1	33.8–34.4	} Yes (ANOVA and Sheffé posttest)	No (t -test)
16–24	5.1	2.8–8.9	17.0	13.3–21.5	11.5	10.7–12.4		
25–34	24.1	18.9–30.1	42.6	37.3–48.1	54.5	53.1–55.9		
35–44	31.2	25.2–37.9	17.3	13.5–21.8	19.6	18.6–20.7		
45–54	22.9	17.7–29.2	15.7	12.2–20.2	9.5	8.7–10.3		
≥55	16.7	13.0–21.3	7.4	5.0–10.8	4.9	4.3–5.5		
Student Status								
Student	14.0	11.3–17.2	NA	NA	14.0	13.1–15.0	} No (χ^2)	NA
Not student	86.0	83.1–89.6	NA	NA	86.0	85.0–86.9		
Household Income								
Mean	93,175	88,660–97,689	NA	NA	81,923	80,813–83,033	Yes (t -test)	NA
<\$50,000	17.9	13.5–23.5	NA	NA	24.4	23.3–25.6	} Yes (χ^2)	NA
\$50,000 < \$75,000	12.5	9.0–17.1	NA	NA	20.3	19.2–21.4		
\$75,000 < \$100,000	16.1	12.1–21.2	NA	NA	16.1	15.1–17.1		
≥\$100,000	53.4	46.8–59.9	NA	NA	39.3	37.9–40.6		
Car–SUV Ownership								
Yes	82.9	77.3–87.3	NA	NA	52.6	46.0–48.7	} Yes (χ^2)	NA
No	17.1	12.7–22.7	NA	NA	47.4	51.3–54.0		
Bicycle Ownership								
Yes	94.2	93.0–97.9	NA	NA	29.4	28.2–30.6	} Yes (χ^2)	NA
No	5.8	3.1–9.0	NA	NA	70.6	69.4–71.8		
Trip Purpose								
Social–personal	32.2	26.6–38.4	22.8	18.6–27.6	44.4	43.0–45.7	} Yes, area cyclists vs. CaBi members only (χ^2)	NA
Recreational–fitness	18.3	14.1–23.5	15.0	11.5–19.2	7.1	6.4–7.8		
Tourism	NA	NA	53.3	47.9–58.6	NA	NA		
Work–meeting	33.2	27.8–39.2	2.4	1.2–4.7	43.2	41.8–44.6		
Other	16.2	12.2–21.7	6.6	4.4–9.8	5.4	4.8–6.0		
Alternative Mode (if not CaBi)								
Induced	NA	NA	2.4	1.2–4.7	3.5	3.0–4.0	} NA	Yes (χ^2)
Walk	NA	NA	52.7	47.3–58.0	30.9	29.7–32.8		
Own bike	NA	NA	1.5	0.6–3.5	5.8	5.2–6.5		
Public transport	NA	NA	35.1	30.1–40.4	45.4	44.1–46.8		
Car	NA	NA	0.9	0.3–2.8	7.2	6.5–7.9		
Taxi	NA	NA	5.4	3.4–8.4	5.8	5.2–6.4		
Other	NA	NA	2.1	1.0–4.3	1.4	1.1–1.7		
Helmet Use								
Yes	NA	NA	6.0	3.9–9.1	36.7	35.4–38.0	} Yes (χ^2)	NA
No	NA	NA	94.1	90.9–96.1	63.3	62.0–64.6		

NOTE: CI = confidence interval; vs. = versus; ANOVA = analysis of variance; NA = not available.

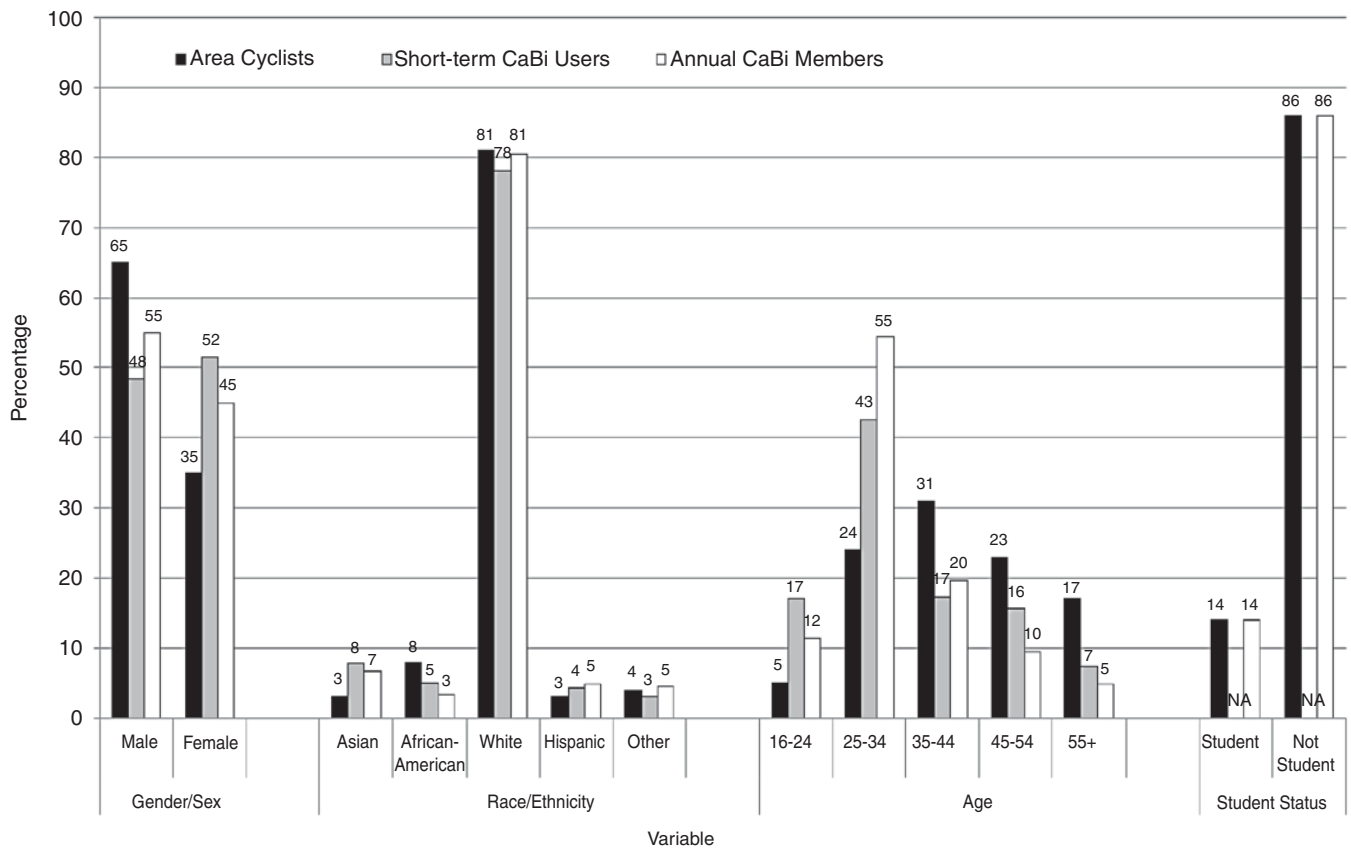


FIGURE 1 Comparison of demographics of Washington, D.C., area cyclists and short-term and annual CaBi members (data on student status not available for short-term CaBi members). (Source: Authors' analysis of CaBi Membership Survey 2011, CaBi Casual User Survey 2011, and MWCOC Travel Survey 2007–2008.)

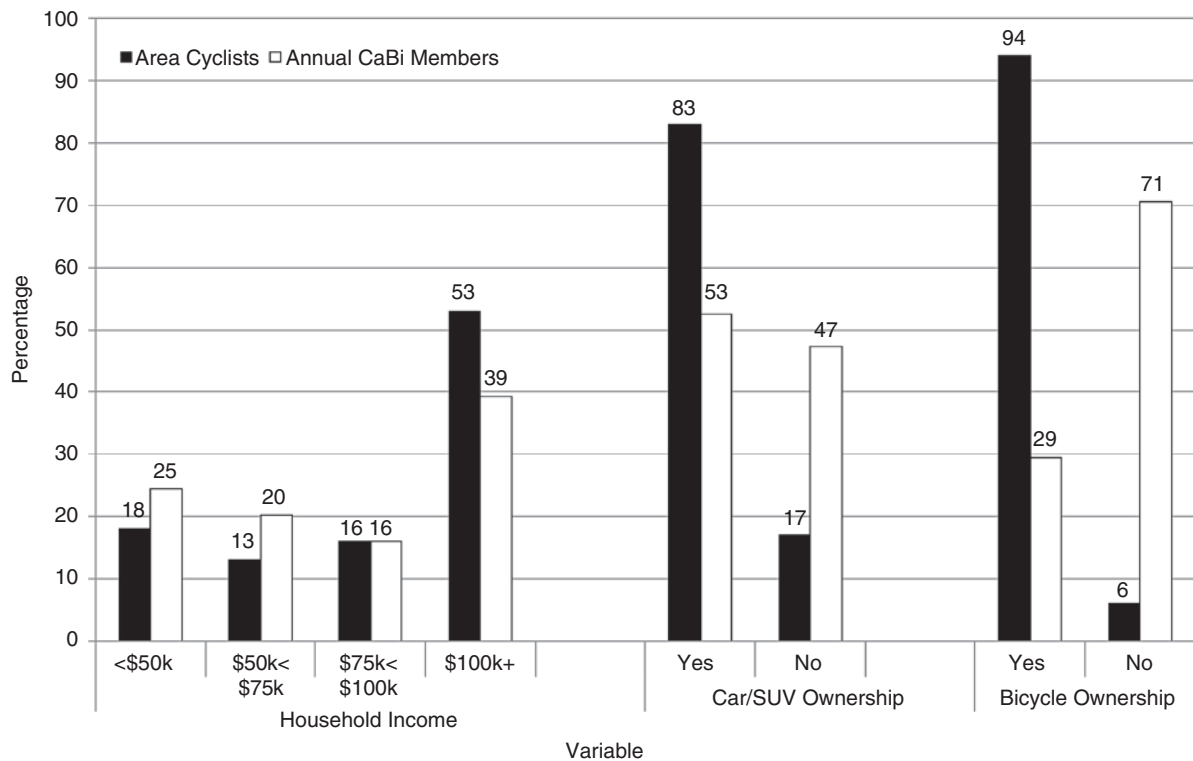


FIGURE 2 Comparison of socioeconomic characteristics of Washington, D.C., area cyclists and annual CaBi members (data not available for short-term CaBi members (k = thousand)). (Source: Authors' analysis of CaBi Membership Survey 2011 and MWCOC Travel Survey 2007–2008.)

was 14% higher than it was for annual CaBi members: \$93,180 compared with \$81,920. The difference in age between area cyclists and CaBi members, reported in the previous section, may help explain dissimilarities in income levels: most CaBi members were younger than 35 years old and had not reached their peak earning years, whereas most area cyclists were older than 35 years old, an age group with comparatively higher incomes.

Area cyclists were more likely to own an automobile than were annual CaBi members. Among D.C. area bicyclists, 83% reported ownership of an automobile, while 17% reported no ownership. A significantly lower rate of automobile ownership was observed among CaBi members of only 53%. Lower car ownership levels among CaBi users might have correlated with their lower incomes and younger ages. Moreover, CaBi users were more likely to live in areas in which CaBi stations were clustered, in the center of Washington, D.C., and along the Rosslyn–Ballston corridor in Arlington County, which had lower automobile ownership rates than outlying areas of Washington, D.C., Arlington, and Alexandria.

Similarly, area cyclists were more likely to own a bicycle: 94% reported ownership of one. The remaining cyclists were assumed to have rented a bicycle or to have borrowed one from a friend or family member. By contrast, only 29% of CaBi members reported ownership of a bicycle, while 71% reported no ownership. The membership survey found that CaBi members that were more than 35 years of age and had incomes of more than \$100,000 were significantly more likely to own both cars and bicycles.

Trip Purpose, Mode Shift, and Helmet Use

Most CaBi short-term users (53%) reported “tourism” as the reason for their trip (Figure 3). Tourism was not an option available to describe trip purpose on either the survey of the D.C. area bicyclists or the survey of CaBi members. “Personal” (44%) and “work” (43%) were cited as the most common trip purposes by CaBi members. D.C. area bicyclists also reported these two purposes as the most common reasons to make a trip by bicycle, although at lower rates than by CaBi members. Reported shares for utilitarian trip purposes were consistent with earlier findings; these earlier findings showed that bicyclists in the Washington, D.C., area rode for utilitarian purposes at high rates. The 88% (44.4% + 43.2% from Table 1 and rounding up) share of trips by CaBi annual members for utilitarian purposes significantly exceeded the 65% of trips by D.C. area bicyclists for utilitarian purposes, the 58% of such trips reported by bicyclists in the entire metropolitan Washington D.C. area, and the 35% reported by bicyclists in all U.S. urbanized areas (18). The CaBi annual member survey results noted that trip purpose was one of the only variables that changed over time. Reported trips became less utilitarian the later the date that membership started. This variable was one of the few in the survey that revealed significant differences between early and late joining members (9).

The mode of transportation replaced by the bicycle trip was not collected by the D.C. area bicyclist survey. Among CaBi users, both short-term users and annual members reported that their bike rides mainly replaced public transport trips (35% and

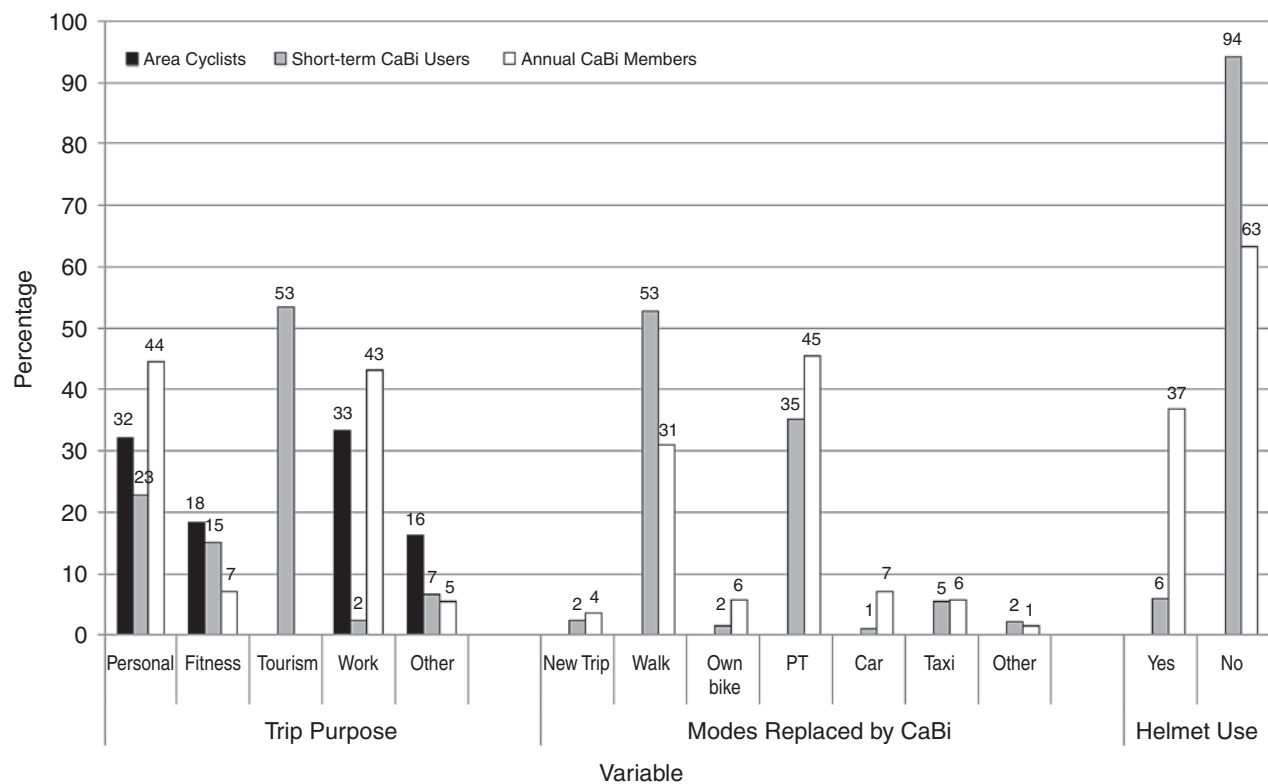


FIGURE 3 Comparison of trip purpose, transport modes replaced by Capital Bikeshare, and helmet use (data on modes replaced and helmet use not available for D.C. area cyclists). (Source: Authors' analysis of CaBi Membership Survey 2011, CaBi Casual User Survey 2011, and MWCOG Travel Survey 2007–2008.)

45%, respectively) and walk trips (53% and 31%, respectively). Short-term users reported significantly lower rates of automobile replacement and personal bicycle trips than annual members. This difference may be explained in part by the fact that most short-term users were tourists, who might not have had access to a personal automobile or bicycle in Washington, D.C. Very few short-term users (2%) and CaBi members (4%) report induced (newly generated) trips, which they would not have made without CaBi.

Finally, data on helmet use were not collected for D.C. area bicyclists. Among CaBi users, 6% of short-term users surveyed were observed to use a helmet, while 37% of annual members reportedly did. The unavailability of helmets to short-term users might have contributed to this difference, and so might have the difference in survey methodology (i.e., observed versus self-reported). More detailed analysis of the annual member survey showed an important difference in helmet use by gender: 48% of male CaBi members reportedly never wore a helmet, compared with only 36% of women. By contrast, 42% of women reported that they always, or almost always, wore a helmet, compared with 31% of men.

FURTHER RESEARCH

This analysis relied on a bivariate comparison of independent variables and bicycling levels, and did not control for other variables. Further analysis could help determine whether or not there were significant differences between CaBi users and D.C. area bicyclists when other factors were controlled for. For example, the lower rate of car ownership and lower age of CaBi riders may correlate with their lower reported household income.

The analysis reported here could not fully consider the spatial restrictions of the two CaBi surveys. Although CaBi covers a substantial portion of the geographic area that the D.C. area bicyclist survey encompassed, the bikesharing system does not serve some outlying areas of Washington, D.C., Arlington County, and Alexandria. Moreover, the CaBi short-term user survey sampled from an even narrower geographic area and focused only on docking stations that experienced a high frequency of short-term rentals, all of which were located within downtown Washington, D.C. Further research should attempt to more precisely control for geography to determine if significant differences persist.

CaBi annual member survey respondents self-selected from e-mail and online solicitations, and responded over the Internet. With such potential biases in this sample, further research that made use of a more randomized sample might yield differing and statistically more representative results, and might be more comparable to the sampling method used for the MWCOG survey.

As noted earlier, the number of short-term CaBi members dramatically increased in 2012. This development may be explained in part by the placement of new docking stations within the National Mall, a popular destination for tourists. A repeat of the CaBi short-term user survey at these sites might yield more insights into the demographics and behavior of CaBi short-term users.

The MWCOG 2007–2008 travel survey provides the latest available data on bicycle use for all trip purposes in metropolitan Washington. The survey provides a snapshot of area cyclists in 2007–2008. The survey does not provide a comprehensive demographic profile of area bicyclists, however, because CaBi began operations in 2010.

Thus, a more recent survey of regular bicyclists may yield insight into how the demographics of regular bicyclists have changed since CaBi began operations.

Finally, results from the Washington, D.C., area cannot be transferred easily to other cities in the United States. Washington is home to the federal government and related industries (e.g., federal contractors, lobbyists), and the area's demographics differ substantially from the national averages. The D.C. area also attracts many tourists from both the United States and abroad. Future studies should compare area bicyclists and bikeshare users in other U.S. cities to test if the differences observed in this study are also found in other places.

CONCLUSIONS

The results of this study suggested that CaBi short-term users and members were different from other Washington, D.C., area cyclists and were more likely to be women, to be younger, to have lower incomes, and to be less likely to own bicycles or automobiles. CaBi users also were more likely to make utilitarian trips than other cyclists. The study indicated that, through the implementation of bikesharing programs, municipalities could encourage new segments of society to cycle and potentially increase the bicycling mode share overall. Given its younger demographic, bikesharing systems could serve as a gateway to future growth of bicycling transportation overall. Decision makers that use bikesharing as a policy to shift transportation mode choice should also be cognizant of other bike-friendly transport policies, such as networks of bike lanes and bike paths.

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